Step 0: Quick Pre-Flight Check (Both Servers Active)

Ensure both the backend and frontend are running:

1. **Frontend**: http://localhost:5173 *(currently running)*
2. **Backend**: `http://localhost:8000` *(currently running with YOLO weights pre-loaded)*

Best Sample Images for the Live Upload Demo

Keep these files ready on your desktop or easily accessible in your file picker:

Recommended Image	Location in Project	What It Demonstrates to Judges
`vid_000052_frame0000070.jpg` ⭐ *(Top Pick)*	backend/samples/	Clear submerged canister/bottle (**Debris 0.75**) right next to an ROV manipulator arm — shows that **ROV equipment is ignored** while the debris is detected.
`vid_000047_frame0000001.jpg`	backend/samples/	High-confidence derelict debris (**Debris 0.81**) on sandy silt.
`vid_000029_frame0000013.jpg`	backend/samples/	**Ecology Honesty Rule**: Detects both a marine animal (**Marine life 0.63**) and plastic (**Debris 0.60**). Proves AquaTriage protects sea life and only targets trash!

The Winning 4-Minute Presentation Flow

mermaid

```
flowchart LR
    A["1. The Hook (Landing Page)"] --> B["2. Command Center & Benchmark"]
    B --> C["3. Live Image Upload & Dehazing"]
    C --> D["4. 2-Opt Route & Printable Dive Plan"]
```

Minute 1: The Problem & The Hook (Start on the Landing Page)

1. Open **http://localhost:5173** in full screen (`F11`).
2. **Say to the judges**:

> *"70% of ocean plastic sinks beyond satellite reach. Today, harbor and salvage teams conduct 'blind lawnmower sweeps'—wasting up to 85% of diver air and ROV battery searching empty silt. AquaTriage turns turbid seabed footage into an optimized extraction mission."*

3. Scroll down briefly to show the **Floating Bathymetric Depth HUD** and the **Glassmorphism Scrollytelling** comparing the 3,030 m blind sweep vs. our 466 m 2-Opt path.
4. Click **"Launch Command Center"** to enter the live workspace.

Minute 2: Show the Multi-Frame Benchmark Survey

1. Click **"Survey 1"** on the left panel.
2. Walk the judges through the interface:
   - **Center Panel**: Show the 8 frames captured by the ROV track. Click through a few scrubber dots.
   - **Tab 3 • Route**: Switch to the **Route** tab to show the seafloor grid. Point out the entry buoy at `(40, 40)` and the cyan waypoints.
   - **Key Stat**: Point out **"84.6% distance saved"** (3,030.6 m reduced down to 466.6 m).

Minute 3: The Live Image Upload (The "Wow" Factor)

1. On the left panel under **UPLOAD TARGET FEED**, click **"Browse" / "Change file"**.
2. Select `vid_000052_frame0000070.jpg` from `backend/samples/`.
3. Highlight the controls:
   - **Detection sensitivity**: Slider at `0.40` – `0.75`.
   - **Optical correction**: Toggle this ON to explain:

> *"Underwater optics suffer from heavy red-wavelength attenuation and turbidity. Our OpenCV pipeline applies Gray-world white balancing and CLAHE in the LAB color space before passing tensors to the model."*

4. Click the vibrant **"RUN DETECTION"** button!
5. **Watch the live inference execute**:
   - The image immediately renders with a yellow/cyan bounding box: `Debris 0.75`.
   - A new pin is calculated and projected onto the seafloor coordinates.
   - The route and pin summary cards update instantly on the right panel.

Minute 4: Scientific Honesty & The Printable Dive Plan

1. **Explain the Ecology Filter**:

> *"Notice how the ROV mechanical arm in the frame was automatically ignored, preventing false positives. If marine life (fish, starfish, crabs) is present, it is labeled in neutral grey and never routed for extraction."*

2. Press `Ctrl + P` (Print Preview):
    - Show the clean **Printable Dive Plan** generated directly from the live data.
    - Explain that field pilots and commercial divers take this laminated checklist onto the boat to execute the salvage run.

Quick Cheat-Sheet for Judge Questions

Question Judges Might Ask	Winning Answer
"What model are you running?"	*"We run an optimized YOLOv8 model trained on MBARI FathomNet and TrashCan 1.0 underwater datasets, specifically fine-tuned for submerged marine debris."*
"How are the 2D coordinates calculated?"	*"Each frame is anchored to the ROV's dead-reckoning trajectory on a 700×460 unit seafloor grid (1 unit = 0.5 m), with local bbox centroid offsets mapped to ground coordinates."*
"How does the route optimization work?"	*"We solve an open Traveling Salesperson Problem (TSP) using a greedy Nearest-Neighbor seed followed by 2-Opt edge swapping, compared against a standard 60-unit boustrophedon sweep."*
"What happens if there's no internet connection offshore?"	*"AquaTriage runs completely local. The FastAPI backend processes everything on-device, and the frontend even includes cached offline fallbacks if the ROV vessel is completely disconnected."*